



SRM VALLIAMMAI ENGINEERING COLLEGE
(An Autonomous Institution)
SRM Nagar, Kattankulathur-603 203



FORMAT FOR STUDENT PROJECT PROPOSAL

1.	Name of the Student (s)	:	Shakthi K / Selvakumar S / Sakthi Velan M
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2.	Name of the Guide	:	Mrs. Priyadharshini M, B.E., M.E., (Ph.D.),
	Department / Designation	:	CSE / Assistant professor
	Institutional Address	:	SRM Nagar, Kattankulathur-603 203
	Phone No. & Mobile No.	:	8939902677
3.	Project Title	:	Immutable Digital Evidence Verification System using Blockchain Technology
4.	Sector in which your Project proposal is to be Considered	:	Engineering & Technology
5.	Project Details	:	1.Introduction In the modern digital era, electronic evidence such as images, videos, documents, and forensic data plays a crucial role in criminal investigations and judicial proceedings. However, digital evidence is highly vulnerable to unauthorized modification, duplication, and tampering during transmission or storage. Traditional centralized evidence management systems lack transparency and integrity assurance, making it difficult to guarantee authenticity throughout the investigation lifecycle. Blockchain technology provides a decentralized and tamper-resistant framework that ensures data immutability and transparency. By integrating blockchain with digital evidence management, it becomes possible to securely store evidence metadata, maintain chain of custody records, and verify evidence integrity at every stage.

		This project proposes an Immutable Digital Evidence Verification System that uses blockchain technology and cryptographic hashing to ensure that digital evidence remains secure, traceable, and legally trustworthy from collection to courtroom presentation.
		2. Objectives The main objectives of the proposed system are: • To develop a secure platform for storing and managing digital evidence. • To ensure integrity verification using cryptographic hash functions. • To implement blockchain technology for immutable evidence tracking. • To maintain a transparent chain of custody among Police, Forensic, and Court authorities. • To detect evidence tampering automatically during verification. • To provide role-based access control for authorized departments. • To enhance trust and transparency in digital forensic investigations.
		3. Methodology The proposed system follows a blockchain-based verification methodology consisting of the following stages: Evidence Collection Police officials upload digital evidence into the system. A unique SHA-256 hash value is generated for the uploaded file, acting as its digital fingerprint.

		Blockchain Recording Evidence metadata and hash values are stored as blocks in a blockchain ledger. Each block is linked with the previous block using cryptographic hashing, ensuring immutability. Chain of Custody Management Every action performed on evidence such as upload, transfer, verification, or approval is recorded permanently in the custody log and blockchain. Forensic Verification Forensic officers download and re-upload the evidence file. The system regenerates the hash and compares it with the original stored hash to confirm integrity. Court Validation The court performs final verification using the same hashing mechanism before approving evidence for legal proceedings. Tamper Detection If any modification occurs, hash mismatch automatically identifies the evidence as tampered.																								
		4. Work Plan <table> <tr> <th>Phase</th><th>Activity</th><th>Duration</th></tr> <tr> <td>Phase 1</td><td>Requirement Analysis & Literature Survey</td><td>Week 1–2</td></tr> <tr> <td>Phase 2</td><td>System Design & Architecture Planning</td><td>Week 3</td></tr> <tr> <td>Phase 3</td><td>Database & Blockchain Module Development</td><td>Week 4–6</td></tr> <tr> <td>Phase 4</td><td>Web Application Development</td><td>Week 7–9</td></tr> <tr> <td>Phase 5</td><td>Evidence Verification & Security Implementation</td><td>Week 10–11</td></tr> <tr> <td>Phase 6</td><td>Testing & Debugging</td><td>Week 12</td></tr> <tr> <td>Phase 7</td><td>Deployment & Documentation</td><td>Week 13–14</td></tr> </table>	Phase	Activity	Duration	Phase 1	Requirement Analysis & Literature Survey	Week 1–2	Phase 2	System Design & Architecture Planning	Week 3	Phase 3	Database & Blockchain Module Development	Week 4–6	Phase 4	Web Application Development	Week 7–9	Phase 5	Evidence Verification & Security Implementation	Week 10–11	Phase 6	Testing & Debugging	Week 12	Phase 7	Deployment & Documentation	Week 13–14
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		5. Budget (Estimated) <table><thead><tr><th>S.No</th><th>Component</th><th>Estimated Cost (₹/Year)</th></tr></thead><tbody><tr><td>1</td><td>Web Hosting</td><td>₹3,000</td></tr><tr><td>2</td><td>Domain Name</td><td>₹900</td></tr><tr><td>3</td><td>SSL Certificate</td><td>₹400</td></tr><tr><td>4</td><td>Cloud Storage</td><td>₹1,200</td></tr><tr><td>5</td><td>Deployment & Maintenance</td><td>₹1,000</td></tr><tr><td colspan="2">Total- ₹6,500/-</td><td></td></tr></tbody></table>	S.No	Component	Estimated Cost (₹/Year)	1	Web Hosting	₹3,000	2	Domain Name	₹900	3	SSL Certificate	₹400	4	Cloud Storage	₹1,200	5	Deployment & Maintenance	₹1,000	Total- ₹6,500/-		
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		6. Any Other Details Technologies Used - Frontend: HTML, CSS, JavaScript - Backend: Python Flask Framework - Database: SQLite - Blockchain: Custom Python Blockchain - Security: SHA-256 Cryptographic Hashing - Deployment: Cloud Platform (Azure / Render) Expected Outcomes - Secure digital evidence storage system. - Immutable blockchain audit trail. - Real-time tampering detection. - Transparent investigation workflow. - Improved reliability of digital evidence in judiciary processes. Future Enhancements - Integration with national forensic databases. - Multi-node blockchain network deployment. - AI-based evidence analysis.																					

			• Cloud storage integration using Azure Blob or AWS S3. • Digital signature and biometric authentication.
7.	Has a similar project been carried out in your college / elsewhere? If so furnish details of the previous project and highlight the improvements suggested in the present one	:	No

CERTIFICATE

This is to certify that Mr. / Miss _____ is a Bonafide final year Student of U.G. Engineering / P.G. professional courses of our college and it is also certified that one copy of utilization certificate and final report will be provided to the institute after completion of the project by the end of April 2026.

Signature of the Guide

Signature of the HOD